

From Competing Pension Regimes to Pillars or Full Funding:

Informality, Fiscal Transition, and Welfare in Colombia

Oliver Pardo*

August 2026

Abstract

This paper studies Colombia's counterfactual transition from the competing pension regimes created by Law 100 of 1993 to two alternatives: the pillar architecture of Law 2381 of 2024 and a fully funded formal system. I embed endogenous informality, PAYG eligibility, portable funded balances, voluntary saving, public debt, and general-equilibrium prices in a two-period overlapping-generations economy. The fiscal block maps Colombia's statutory 55 percent net-debt anchor and 0.2 percent structural primary surplus into a 40-year model period. A smooth consumption wedge and a separate primary-balance adjustment stabilize debt without forcing the consumption tax to absorb each PAYG shock contemporaneously. In the Law 100 steady state, informality is 53.2 percent and funded accounts hold 77.0 percent of household assets. Under pillars, informality rises to 77.9 percent, capital falls 14.1 percent, and the stationary consumption tax is 55.4 percent. Funded saving falls by 70.3 percent, but voluntary saving rises by 173.8 percent. The high tax is therefore not explained by the disappearance of all nonpension saving or by a larger net PAYG deficit; it mainly reflects the endogenous contraction of the consumption-tax base and other fiscal revenue. The transition is damped and converges to the debt anchor. In the fully funded counterfactual, informality falls to 10.2 percent, capital more than doubles, and stationary welfare rises 22.9 percent in consumption-equivalent terms. Its negative 3.0 percent residual fiscal

*Departamento de Administración, Pontificia Universidad Javeriana, Bogotá, Colombia. Email: pardoo@javeriana.edu.co.

wedge is a model-implied dividend, not a proposed negative VAT. These comparisons identify mechanisms rather than provide forecasts: one model period spans a generation, domestic public debt does not crowd out household capital in the central specification, and Law 2381 has not been implemented.

Keywords: pension reform; informality; pillar system; PAYG; funded accounts; public debt; transitional dynamics; distributional welfare. **JEL:** E21; H55; J26; J46.

1 Introduction

Workers in economies with large informal sectors choose both whether to enter formal employment and how much value they obtain from mandatory pension contributions. Under Colombia’s Law 100 of 1993, formal workers could choose between a PAYG regime and an individual funded account ([Congreso de Colombia, 1993](#)). Law 2381 of 2024 replaces that competition with complementarity: every formal worker contributes to PAYG on earnings up to 2.3 statutory minimum wages, while only the excess enters the funded component ([Congreso de Colombia, 2024](#)). The reform therefore changes the choice set itself, not merely contribution rates.

This paper asks three questions. How does that institutional change affect informality, capital, wages, taxes, and pension participation? Which cohorts gain or lose during the transition? How are gains and losses distributed across worker types within each generation? These are counterfactual questions. The Constitutional Court suspended the integral entry into force of Law 2381 while it reviews the corrected legislative procedure ([Corte Constitucional de Colombia, 2025](#)); the experiment below is consequently not an observed before-and-after evaluation.

I answer the questions in a two-period overlapping-generations economy. One period represents roughly 40 years. Heterogeneous workers choose informality or formality, voluntary saving, and, when Law 100 applies, one of two pension regimes. A funded balance is portable and earns the market return. PAYG eligibility increases with worker type; an ineligible contributor receives a refund, while an eligible contributor obtains a defined benefit with a floor and a ceiling. These differences produce the benchmark ordering

$$\text{informal} \longrightarrow \text{capitalization} \longrightarrow \text{PAYG}$$

without quotas or random-utility shocks. Under Law 2381, the second choice disappears.

A formal worker receives a single package containing PAYG and, above the statutory threshold, funded saving. The endogenous allocation is therefore among informality, PAYG-only formality, and mixed PAYG–funded formality.

The legal crosswalk matters quantitatively. The total contribution is 16 percent. Thirteen points on earnings up to 2.3 minimum wages finance the common PAYG fund, and 13.2 points on the excess are credited to the individual account. The PAYG replacement rate is $65.5 - 0.5s$ percent, where s is the covered income in minimum wages, and its contributory benefit cannot fall below one minimum wage. The solidarity benefit is normalized to the 2023 extreme-poverty line, $218,846/1,160,000 = 0.1887$ minimum wages, and its targeted mass is proxied by the 12.6 percent incidence of extreme poverty (Departamento Administrativo Nacional de Estadística, 2025; Presidencia de la Republica de Colombia, 2022).

The fiscal comparison holds net debt at the legal long-run anchor of 55 percent of annual GDP and targets the 0.2 percent structural primary surplus associated with that anchor (Congreso de Colombia, 2021). This is not a description of current debt: the escape clause is active in 2026–2027 and the 2026 Medium-Term Fiscal Framework projects central-government net debt at 58.9 percent of GDP in those years (Ministerio de Hacienda y Credito Publico, 2026). The legal anchor is used only as a stationary normalization.

The Law 100 benchmark has 53.2 percent informality, 43.4 percent funded-only formality, and 3.4 percent PAYG. The central pillar steady state has 77.9 percent informality, 7.0 percent PAYG-only formality, and 15.1 percent mixed formality. Its stationary consumption tax is 55.4 percent. This result is not produced by a larger PAYG cash deficit: the deficit falls from 0.0359 to 0.0017 model units. Instead, funded saving falls from 0.0226 to 0.0067, total capital falls 14.1 percent, and the endogenous rise in informality reduces the consumption-tax base by 44.2 percent and non-consumption revenue by 19.5 percent. Voluntary saving rises sharply, so the mechanism is a change in the composition and level of saving, amplified by the fiscal base.

Both perfect-foresight paths converge. Under pillars, the tax rises smoothly from 22.1 to 55.4 percent; debt peaks at 60.4 percent of annual GDP and returns to 55 percent; and the stationary welfare loss is 3.0 percent. Under the fully funded counterfactual, the full 16 percent formal contribution enters individual accounts, new entrants acquire no PAYG rights, the solidarity benefit remains, and inherited Law 100 obligations are honored. Capital rises from 0.0293 to 0.0611, informality falls to 10.2 percent, and stationary welfare rises 22.9 percent. The counterfactual produces a residual fiscal dividend of 3.0

percent because other revenue exceeds nonpension spending plus the structural primary-surplus target. The paper contributes an endogenous institutional mapping, a transition experiment that does not impose convergence without checking the terminal equations, an explicit debt-and-tax closure, and a distributional accounting of wages and welfare within and across cohorts. The rest of the paper reviews the literature, presents the model, maps both laws into parameters, reports stationary and transition results, and details the algorithm and its validation.

2 Related literature

The model follows the overlapping-generations approach to social security in [Diamond \(1977\)](#). Mandatory retirement programs can address undersaving ([Feldstein, 1985](#)), while present-biased preferences provide a behavioral rationale for mandatory contributions ([Laibson, 1998](#); [İmrohoroglu et al., 2003](#)). [Pardo \(2023\)](#) introduces mandatory retirement saving into an economy with endogenous informality. The present paper adds pension-institution choice under Law 100 and then removes that choice under the mandatory pillar package. This permits a transition analysis in which the policy changes both returns and the feasible alternatives.

Pension design cannot be separated from labor informality. Payroll-financed programs alter the value of formal employment ([Levy, 2008](#)), and informality reflects choices by workers and firms rather than a residual labor-market state ([Ulyssea, 2018](#)). Search frictions and human-capital accumulation create heterogeneous formal careers ([Bobba et al., 2021](#)). The shadow economy also changes the information and tax bases available for redistribution ([Doligalski and Rojas, 2023](#)).

Quantitative pension models with informality include [Joubert \(2015\)](#) and [McKiernan \(2020\)](#). Related work studies retirement and occupational behavior when formal contributions can be avoided ([Ferreira and Parente, 2020](#); [Moreno, 2022](#)). The mechanism studied here is the interaction between pension eligibility, contribution portability, and the formal-informal margin. The stationary benchmark identifies conditions for interior regime coexistence; the reform experiment then studies how mandatory complementarity propagates through fiscal closure, wages, and cohort welfare. The debt experiment shows that the oscillation under strict contemporaneous balance is not a structural prediction of pension design: it depends on whether the fiscal authority can transfer a bounded part of the adjustment across cohorts.

For Colombia, [Becerra \(2025\)](#) develops the CEDE Pension Model, a dynamic mi-

crossimulation that combines household surveys, administrative records, legal rules, and population projections. That model supplies the institutional rates and empirical anchors used below. Its annual partial-equilibrium accounting and the present model's 40-year general-equilibrium period are not interchangeable. The calibration therefore uses a documented crosswalk rather than copying the entire parameter vector.

3 Model

3.1 Demography, heterogeneity, and preferences

One model period represents approximately 40 years. Population grows at rate n , labor-augmenting productivity at rate g , and a worker survives to old age with probability $1 - m$. Type $i \in [0, 1]$ has density

$$f(i) = \frac{2.4450}{1.006341403957648} \exp \left[- \left(\frac{i - 0.4655}{0.2329} \right)^2 \right], \quad \int_0^1 f(i) di = 1. \quad (1)$$

Sector-specific efficiency is

$$z_f(i) = \text{base}_f e^{\rho_f i}, \quad z_s(i) = \text{base}_s e^{\rho_s i}. \quad (2)$$

Instantaneous utility is CRRA, $u(c) = c^{1-\sigma}/(1-\sigma)$. Decision utility and experienced utility are

$$V_j(i) = u(c_{1j}(i)) + \beta(1 - m)\delta u(\tilde{c}_{2j}(i)), \quad (3)$$

$$W_j(i) = u(c_{1j}(i)) + (1 - m)\delta u(\tilde{c}_{2j}(i)), \quad (4)$$

where $j \in \{I, C, P\}$ denotes informal employment, capitalization, and PAYG. The parameter $\beta < 1$ affects saving and regime choice but is not used as a normative welfare weight.

3.2 Technology and household budgets

Formal and informal output are

$$Y_f = K^\alpha N_f^{1-\alpha}, \quad Y_s = A_s N_s. \quad (5)$$

Competitive pre-tax prices satisfy

$$\tilde{r} + \delta_k = \alpha(N_f/K)^{1-\alpha}, \quad \tilde{w}_f = (1 - \alpha)(K/N_f)^\alpha, \quad \tilde{w}_s = A_s. \quad (6)$$

Taxes map these into the net return r and the effective wages w_f and w_s .

For choice j , voluntary saving $a_j(i) \geq 0$ satisfies

$$(1 + \tau_c)c_{1j}(i) + (1 + g)a_j(i) = y_{1j}(i), \quad (7)$$

$$(1 - m)(1 + \tau_c)c_{2j}(i) = (1 + r)a_j(i) + p_j(i). \quad (8)$$

Informal income is $y_{1I}(i) = w_s z_s(i)$. Both formal alternatives have

$$y_{1C}(i) = y_{1P}(i) = (1 - \tau_\ell - \tau_p) w_f z_f(i). \quad (9)$$

Both formal alternatives pay the statutory rate τ_p . The funded account credits $\tau_C \leq \tau_p$, while the PAYG notional balance credits $\tau_N \leq \tau_p$; uncredited amounts enter consolidated public revenue. The regimes therefore share the current wedge but differ in credited balances and retirement benefits. Baseline transfers are zero.

3.3 Capitalization

A funded contribution is fully vested and portable:

$$A_C(i) = \tau_C w_f z_f(i), \quad (10)$$

$$p_C(i) = \frac{(1 + r)A_C(i)}{1 + g}. \quad (11)$$

The contribution adds to the capital stock. The worker receives the accumulated value regardless of whether the subsequent formal career would have satisfied PAYG vesting requirements.

3.4 PAYG eligibility, refund, and pension formula

The probability that type i completes the PAYG eligibility requirement is

$$q(i) = \frac{\exp(\gamma_0 + \gamma_1 i)}{1 + \exp(\gamma_0 + \gamma_1 i)}, \quad \gamma_1 > 0. \quad (12)$$

The pension conditional on eligibility is

$$\mathcal{B}(y_i) = \min \{B_{\max}, \chi \max [B_{\min}, by_i]\}, \quad y_i = w_f z_f(i), \quad (13)$$

where b is the replacement parameter and χ converts the replacement flow into units of the long model period. It also captures the implicit subsidy embedded in the PAYG formula.

If the worker fails to qualify, a fraction φ of the notional balance is returned. Let r_N denote the annual real return credited to that balance. Expected PAYG retirement income is therefore

$$p_P(i) = q(i) \frac{(1-m)\mathcal{B}(y_i)}{1+g} + [1-q(i)]\varphi \frac{(1+r_N)^{40}\tau_N y_i}{1+g}. \quad (14)$$

In the Colombian calibration, $\varphi = 1$ and $r_N = 0$: a non-eligible worker recovers the notional contribution adjusted for inflation but receives no real return. Expected PAYG benefits, including refunds, are public expenditure.

3.5 Joint choice and endogenous thresholds

Every worker solves the saving problem under all three alternatives and chooses

$$j^*(i) = \arg \max_{j \in \{I, C, P\}} V_j(i). \quad (15)$$

An interior three-regime equilibrium has thresholds $0 < i_1 < i_2 < 1$ satisfying

$$V_I(i_1) = V_C(i_1) > V_P(i_1), \quad (16)$$

$$V_C(i_2) = V_P(i_2) > V_I(i_2). \quad (17)$$

The resulting allocation is

$$j^*(i) = \begin{cases} I, & i < i_1, \\ C, & i_1 \leq i < i_2, \\ P, & i \geq i_2. \end{cases} \quad (18)$$

Equation (18) is an equilibrium outcome. The algorithm evaluates all three utilities point by point and does not impose either threshold.

3.6 The pillar reform

Law 2381 changes both the alternatives and the contribution bases. Let $w_{\min}$ denote one minimum wage in model units and $\bar{y} = 2.3w_{\min}$. For a formal worker with earnings y_i , the PAYG and funded bases are

$$y_i^P = \min\{y_i, \bar{y}\}, \quad (19)$$

$$y_i^C = \max\{y_i - \bar{y}, 0\}. \quad (20)$$

The worker pays the 16 percent total contribution on both bases. Thirteen percent of y_i^P enters the common PAYG fund; 13.2 percent of y_i^C is credited to the individual account. The remaining shares finance administration, insurance, and the public savings fund, which are consolidated in the government budget.

For covered earnings measured in minimum wages, $s_i = y_i^P/w_{\min}$, the statutory PAYG rate is

$$b^{2381}(s_i) = 0.655 - 0.005s_i. \quad (21)$$

The modeled defined benefit is

$$\mathcal{B}^{2381}(y_i^P) = \chi_{2381} \max\{w_{\min}, b^{2381}(s_i)y_i^P\}, \quad (22)$$

subject to the statutory maximum. The central experiment sets $\chi_{2381} = 1$ because the long-period conversion is not legally identified; $\chi_{2381} = 2.9$, the value selected for the Law 100 interior branch, is reported only as a stress test.

The solidarity transfer is targeted to the lowest types among informal workers. With cutoff i_S chosen so that $F(i_S) = 0.126$,

$$S(i) = \mathbf{1}\{j^*(i) = I, i \leq i_S\}, \quad B^S = 0.1887w_{\min}. \quad (23)$$

This is a parsimonious proxy for poverty targeting. It does not separately represent the semicontributory pillar, contribution histories, disability, sex-specific ages, or household means tests.

Formal workers now choose a package rather than an exclusive pension regime:

$$j_{2381}^*(i) = \arg \max_{j \in \{I, F\}} V_j(i). \quad (24)$$

A formal worker with $y_i \leq \bar{y}$ is classified as PAYG-only, while one with $y_i > \bar{y}$ is classified

as mixed PAYG-funded. Thus positive masses in the three reported categories arise from the endogenous formality threshold and the statutory earnings threshold; they are not three pension-plan choices under the new law.

3.7 Aggregation and fiscal closure

Let $d_j(i) = \mathbf{1}\{j^*(i) = j\}$. Regime shares and effective labor are

$$\pi_j = \int_0^1 d_j(i) f(i) di, \quad (25)$$

$$N_f = \int_0^1 [d_C(i) + d_P(i)] z_f(i) f(i) di, \quad (26)$$

$$N_s = \int_0^1 d_I(i) z_s(i) f(i) di. \quad (27)$$

Gross household assets carried to the next period are

$$S' = \frac{1}{1+n} \int_0^1 a_{j^*(i)}(i) f(i) di + \frac{1}{(1+n)(1+g)} \int_0^1 d_C(i) \tau_C w_f z_f(i) f(i) di. \quad (28)$$

Government bonds and productive capital may be imperfectly integrated. If B' is public debt per efficiency unit of the incoming cohort and ω_B is the share absorbed from the modeled household portfolio, asset-market clearing is

$$K' = S' - \omega_B B'. \quad (29)$$

The central experiment sets $\omega_B = 0$: public debt is a fiscal liability held outside the modeled household portfolio. This avoids imposing mechanical crowding out over a 40-year period, but it is an identifying assumption rather than an empirical estimate.

Public pension revenue under Law 100 is

$$R^{pen} = \int_0^1 [d_P(i) \tau_p + d_C(i) (\tau_p - \tau_C)] w_f z_f(i) f(i) di. \quad (30)$$

Write R_t^{nc} for labor, payroll, capital, and public pension revenue; E_t for spending, transfers, solidarity benefits, and PAYG obligations; and $\mathcal{C}_t$ for the consumption-tax base. The raw primary balance is

$$PB_t^{raw} = \tau_{ct} \mathcal{C}_t + R_t^{nc} - E_t. \quad (31)$$

Let $H = 40$ be years per model period, $G = (1 + n)(1 + g)$, and Y total output in model units. The statutory net-debt anchor and structural primary-surplus target are mapped as

$$\bar{B} = \frac{0.55\bar{Y}}{H}, \quad \overline{PB} = 0.002\bar{Y}. \quad (32)$$

The 71 percent debt limit is a legal ceiling, not the stationary target. To make (32) stationary, the gross long-period return on public debt is $R_B = G + \overline{PB}/\bar{B}$. Public debt then obeys

$$GB_{t+1} = R_B B_t - PB_t. \quad (33)$$

This public-debt return is distinct from the return on productive capital.

A single consumption tax that closes every contemporaneous fiscal gap is locally unstable in this model because a higher tax raises informality and erodes its own base. I therefore use a two-instrument transition closure. Define

$$z(\tau) = \log \left(\frac{\tau - \tau^{min}}{\tau^{max} - \tau} \right). \quad (34)$$

The fiscal wedge approaches its final stationary value smoothly,

$$z(\tau_{ct}) = \rho_\tau z(\tau_{c,t-1}) + (1 - \rho_\tau) z(\bar{\tau}_c), \quad (35)$$

while the primary-balance rule is

$$PB_t^* = \overline{PB} + \phi_B(B_t - \bar{B}), \quad \phi_B = R_B - \rho_B G, \quad (36)$$

$$A_t = PB_t^* - PB_t^{raw}, \quad PB_t = PB_t^{raw} + A_t. \quad (37)$$

A positive A_t is a temporary spending reduction or equivalent non-distortionary revenue; a negative value is temporary fiscal space. It vanishes in each stationary equilibrium. Equation (36) gives debt deviations persistence ρ_B , while (35) prevents the consumption wedge from reproducing the former two-cycle. The lower bound allows a negative residual wedge, interpreted as a dividend or rebate.

Under pillars, the funded contribution in gross asset accumulation is

$$\frac{\int \mathbf{1}\{j^* = F\} 0.132 y_i^C f(i) di}{(1 + n)(1 + g)},$$

and public pension revenue is

$$R_{2381}^{pen} = \int_0^1 \mathbf{1}\{j^*(i) = F\} [0.16y_i^P + (0.16 - 0.132)y_i^C] f(i) di. \quad (38)$$

In the fully funded counterfactual, the full $0.16y_i$ enters the individual account and $R^{pen} = 0$ for new entrants. In both reforms, acquired Law 100 rights of the old cohort enter E_t and are honored during the transition.

4 Calibration

4.1 Parameter crosswalk

The Colombian inputs come from the CEDE Pension Model documented by [Becerra \(2025\)](#). That source is an annual dynamic microsimulation; the present model is a stationary 40-year OLG equilibrium. I therefore classify each input as a direct legal parameter, a comparable calibration moment, an external validation, or a retained structural parameter. This prevents annual accounting assumptions from being reinterpreted as balanced-growth parameters.

For the Law 100 benchmark, the direct legal inputs are the 16 percent total contribution rate, the 11.5 percent credited to an individual account, and the 13 percent credited to the PAYG notional balance. A funded worker pays the full 16 percent: the credited balance becomes capital and the remainder enters consolidated public revenue. A PAYG worker also pays 16 percent, while a failed eligibility draw returns the 13 percent notional balance with no real interest. The replacement parameter is 0.65 near one minimum wage.

The reform crosswalk is separate. Under Law 2381, 13 points below the 2.3-minimum-wage threshold enter the common PAYG fund, while 13.2 points above it enter the funded account. The remaining 2.8 points on the excess consolidate administration, insurance, and public saving. The legal replacement formula and one-minimum-wage floor enter directly. The solidarity amount uses the 2023 extreme-poverty line, while the share of low types eligible for targeting uses the national 12.6 percent extreme-poverty incidence. These are a benefit ratio and a targeting proxy, respectively; neither should be interpreted as an estimate of take-up among older adults.

Table 1 reports the complete benchmark. Annual population growth, productivity growth, and discounting remain compounded over 40 years: $n = 1.013^{40} - 1$, $g =$

$1.004^{40} - 1$, and $\beta = 0.97^{40}$. The source assumes 1.6 percent annual real wage growth, but that rate only indexes earnings and benefits in its microsimulation. Substituting it for balanced-growth productivity g changes every intergenerational scaling equation and does not yield a stable benchmark. Likewise, the source’s 4 percent real return is retained as an external check rather than imposed on the endogenous return of a closed economy.

Table 1: Baseline parameters and provenance

Parameter	Value	Use	Interpretation
m	0.4000	retained	mortality before old age
n	0.6764	retained	population growth per model period
σ	2.5000	retained	CRRA curvature
β	0.2957	retained	behavioral discount factor
g	0.1731	retained	balanced-growth productivity
α	0.4000	retained	capital share
A_s	0.08215	calibrated	matches the long-run formality anchor
ρ_s, ρ_f	2.90, 4.64	retained	type gradients
τ_ℓ, τ_n	0.125, 0.125	retained	labor and payroll rates
τ_p	0.1600	direct	total pension contribution
τ_C	0.1150	direct	credited funded-account rate
τ_N	0.1300	direct	credited PAYG notional rate
r_N	0.0000	direct	real return on a PAYG refund
τ_k	0.3200	retained	capital-income tax
b	0.6500	direct	replacement rate near one minimum wage
G	0.3300	retained	government purchases
γ_0	-8.7900	calibrated	matches the eligibility proxy
γ_1	15.0000	retained	shape normalization for eligibility
φ	1.0000	direct	refund fraction if not eligible
χ	2.9000	selected	40-year benefit conversion; interior branch
$B_{\min}$	0.0500	retained	floor in internal OLG units
$B_{\max}$	100.0000	retained	nonbinding ceiling

4.2 Moments and external checks

The CEDE projection raises formal employment from 42 percent in 2020 to 48.5 percent in 2100. I use 48.5 percent as the stationary formality anchor. It also projects contributory coverage near 25 percent in 2080–2100. Because the OLG has no contribution-history state, I approximate conditional eligibility among formal workers by $0.25/0.485 = 0.51546$. This ratio is a calibration proxy, not an accounting identity: contributory coverage in the source includes DB, DC, and DB+DC pensions and is measured among older adults.¹

¹The coverage rows in Table 2 of [Becerra \(2025\)](#) are labeled in the reverse order. Section 4.1, Figure 4, and Table 1 consistently report contributory coverage of 23.7–23.8 percent on average and non-contributory coverage of 53.1–53.3 percent. The calibration uses the separately reported long-run contributory value of approximately 25 percent.

Table 2: Law 2381 policy mapping

Parameter	Value	Status
Total contribution rate	0.1600	direct legal input
PAYG credit below threshold	0.1300	direct legal input
Funded credit above threshold	0.1320	direct legal input
PAYG earnings threshold	2.3000 SMLMV	direct legal input
Replacement intercept	0.6550	direct legal input
Replacement slope	0.0050	direct legal input
Minimum contributory benefit	1.0000 SMLMV	direct legal input
Solidarity benefit	0.1887 SMLMV	calibration ratio
Solidarity target mass	0.1260	calibration proxy
χ_{2381} central (stress)	1.0000 (2.9000)	not identified

Notes: SMLMV denotes the statutory monthly minimum wage. The solidarity ratio is 218,846/1,160,000. The 2.9 multiplier is not a legal input.

The two comparable moments discipline A_s and γ_0 . The 10 percent DB enrollment share among young entrants and the 4 percent real return in the source are not imposed. They are external checks because the former is a cohort-specific plan assignment and the latter is an actuarial assumption rather than a general-equilibrium price. Table 3 reports both the matched moments and these transparent discrepancies.

Table 3: Calibration moments and external validation

Use	Moment	Target	Model	Gap
Calibrated	Formal share	0.48500	0.49081	0.00581
Calibrated	Eligibility among formals	0.51546	0.51373	-0.00173
Validation	PAYG share among formals	0.10000	0.06353	-0.03647
Validation	Annual real return	0.04000	0.05741	0.01741

Notes: The eligibility target is the proxy 0.25/0.485. The PAYG validation uses the approximately 10 percent DB enrollment share among entrants aged 20–30 reported for 2017.

The source does not identify the logistic slope, the conversion from an annual pension flow to a 40-year benefit, or the minimum wage in the model’s efficiency-wage units. Accordingly, γ_1 , χ , and the wage normalization are not empirical estimates. The value $\chi = 2.9$ selects the interior Law 100 branch; it is not transferred to the central reform experiment. The reported stationary reform solution is refined to 201 types, while the transition uses 101 diagnostic types plus exact root insertion at every choice boundary.

The fiscal-transition parameters combine legal targets and provisional dynamics. Ar-

ticle 60 of Law 2155 sets a 55 percent net-debt anchor, a 71 percent limit, and a 0.2 percent structural primary surplus when debt is at the anchor (Congreso de Colombia, 2021). The escape clause does not replace those statutory long-run values (Ministerio de Hacienda y Credito Publico, 2026). The persistence coefficients are reduced-form numerical calibrations. The bounds define the domain of the fiscal wedge, not proposed tax rates.

Table 4: Fiscal targets and transition parameters

Parameter	Interpretation	Value
H	Years per model period	40
$H\bar{B}/\bar{Y}$	Net-debt anchor / annual GDP	0.55
$\overline{PB}/\bar{Y}$	Structural primary surplus	0.002
ρ_B	Debt-deviation persistence	0.25
ρ_τ	Fiscal-wedge persistence	0.70
ω_B	Domestic portfolio absorption	0
τ^{min}	Lower fiscal-wedge bound	-0.20
τ^{max}	Upper fiscal-wedge bound	0.90

Notes: The debt anchor and primary-balance target are legal inputs. Persistence parameters, domestic absorption, and wedge bounds are modeling assumptions. Current Colombian debt need not equal the statutory long-run anchor.

5 Results

5.1 Law 100 benchmark

Table 5 reports the refined Law 100 stationary equilibrium with the fiscal anchor imposed. All three alternatives have positive mass. Informality is 53.2 percent, funded-only formality 43.4 percent, and PAYG 3.4 percent. The formal-entry threshold is $i_1 = 0.4790$ and the pension-regime threshold is $i_2 = 0.7641$.

The first threshold reflects the formal tax wedge. Types below 0.4790 obtain too little formal-sector income to compensate for taxes and pension contributions. Above it, capitalization is attractive because the funded balance is portable. At $i = 0.7641$, expected PAYG eligibility and the implicit subsidy overtake the funded return. The three-way ordering therefore remains plausible after introducing the legal debt anchor.

Table 5: Endogenous-choice Law 100 steady state

Outcome	Value	Outcome	Value
Informal share	0.53182	Capital stock	0.02932
Funded-only share	0.43373	Formal labor	8.68561
PAYG share	0.03444	Informal labor	1.49887
First threshold i_1	0.47898	Consumption tax	0.22137
Second threshold i_2	0.76407	Annualized return	0.05725
Young consumption	0.39060	Formal wage	0.05474
Old-age consumption	0.26366	Voluntary capital	0.00675
Funded capital	0.02256	PAYG contributions	0.01434
Decision welfare	-5.46629	PAYG benefits	0.05023
Experienced welfare	-5.97437	Maximum residual	3.94×10^{-6}

5.2 Stationary pillars and full funding

Table 6 compares the three stationary allocations. Under Law 2381, formal workers cannot select an exclusive funded regime: 7.0 percent are PAYG-only and 15.1 percent are mixed. Informality rises to 77.9 percent. The fully funded counterfactual sends the entire 16 percent formal contribution to an individual account, preserves the solidarity benefit, and creates no PAYG rights for new entrants. It produces 89.8 percent funded-only formality and 10.2 percent informality.

The pillar result refines the conjecture that the reform eliminates saving. Funded capital falls 70.3 percent and its share of household assets falls from 77.0 to 26.6 percent, but voluntary capital rises 173.8 percent. Total capital falls 14.1 percent rather than vanishing. Nor does a larger net PAYG deficit explain the 55.4 percent tax: the cash deficit shrinks from 0.0359 to 0.0017 model units. Government spending falls only 1.2 percent, while non-consumption revenue falls 19.5 percent and the consumption-tax base falls 44.2 percent. The high stationary tax is mainly the general-equilibrium consequence of the endogenous informality–tax-base feedback, reinforced by the loss of funded capital.

The stress experiment transfers the underidentified $\chi = 2.9$ from Law 100 sorting to the pillar benefit. Informality becomes 79.5 percent and the stationary tax 77.7 percent. I report it as a warning against treating the multiplier as a portable legal parameter, not as the central implementation of Law 2381.

The fully funded result operates in the opposite direction. Funded capital nearly triples and output rises 39.6 percent. Broader formality raises other revenue above non-pension spending plus the structural primary-surplus target. The resulting 3.0 percent

Table 6: Stationary comparison under the common fiscal anchor

Outcome	Law 100	Pillars	Fully funded
Informal share	0.5318	0.7795	0.1016
Funded-only share	0.4337	0.0000	0.8984
PAYG-only share	0.0344	0.0697	0.0000
Mixed PAYG–funded share	0.0000	0.1508	0.0000
Fiscal wedge	0.2214	0.5544	−0.0298
Capital stock	0.02932	0.02518	0.06113
Output	1.01463	0.87164	1.41651
Voluntary capital	0.00675	0.01849	0.00029
Funded capital	0.02256	0.00669	0.06084
Funded share of household assets	0.7696	0.2657	0.9953
Formal wage	0.05474	0.06090	0.06606
PAYG cash balance	−0.03589	−0.00172	0.00000
Net debt / annual GDP	0.5500	0.5500	0.5500
Primary balance / GDP	0.0020	0.0020	0.0020
Stationary welfare CEV	0.0000	−0.0299	0.2289

Notes: CEV is the aggregate consumption-equivalent change relative to the Law 100 allocation. The negative fully funded fiscal wedge is a residual dividend or rebate in the model’s consolidated budget, not a proposed negative VAT.

fiscal dividend should not be treated as a calibrated policy recommendation. It depends on the central assumption $\omega_B = 0$, on a 40-year model period, and on the absence of administrative costs, annuity-market frictions, and transition compensation beyond acquired rights.

5.3 Transition to the pillar system

The reform begins with cohort 1. Those old at the announcement date retain Law 100 contributory rights. New entrants face the pillar rules. Figure 1 shows the complete 20-cohort path.

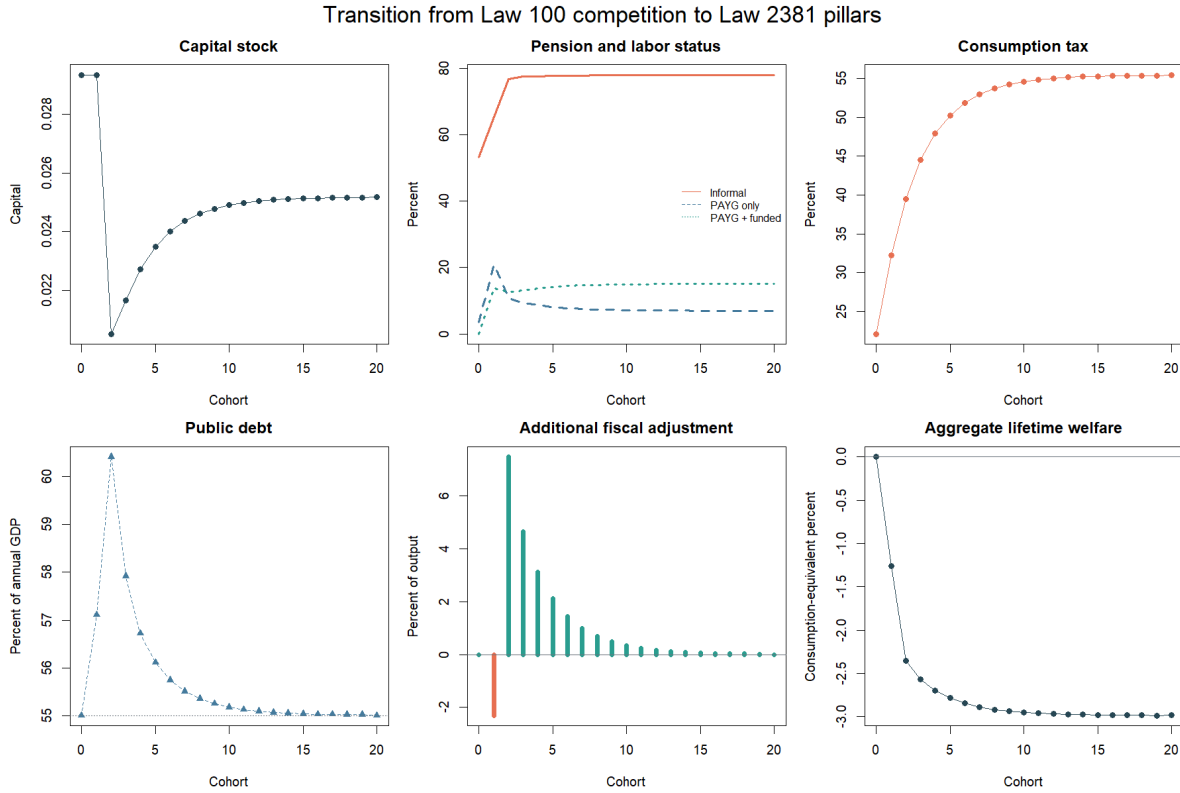


Figure 1: Transition from Law 100 to the pillar system

Notes: One cohort is one model period of approximately 40 years. Public debt is measured as a share of annual GDP; the horizontal reference is the 55 percent legal anchor. The additional fiscal adjustment is $100A_t/Y_t$ and is zero in both endpoints.

The path no longer has the alternating “accordion” pattern. The fiscal wedge rises from 22.1 percent to 32.3 percent for cohort 1, 39.5 percent for cohort 2, and 55.4 percent in the limit. Informality rises in the same damped direction. Capital falls sharply when the first new pension balances mature, reaches 0.0205 for cohort 2, and converges from

below to 0.0252. Debt initially overshoots, peaking at 60.4 percent of annual GDP for cohort 2, and then returns to 55 percent.

Table 7: Selected points: transition to pillars

Outcome	Law 100	Cohort 1	Cohort 2	Cohort 5	Steady state
Informal share	0.5318	0.6538	0.7685	0.7772	0.7795
PAYG-only share	0.0344	0.2082	0.1074	0.0811	0.0697
Mixed share	0.0000	0.1380	0.1240	0.1417	0.1508
Fiscal wedge	0.2214	0.3227	0.3951	0.5024	0.5544
Capital stock	0.02932	0.02932	0.02054	0.02349	0.02518
Debt / annual GDP	0.5500	0.5711	0.6041	0.5611	0.5500
Additional adjustment / output (percent)	0.00	-2.31	7.47	2.12	0.00
Aggregate welfare CEV (percent)	0.00	-1.26	-2.36	-2.79	-2.99

The additional fiscal instrument changes sign once: cohort 1 uses temporary fiscal space of 2.3 percent of output, whereas cohort 2 requires an adjustment of 7.5 percent as inherited obligations come due. The adjustment then decays to zero. The terminal gap is 4.00×10^{-4} , the debt-identity residual is zero to machine precision, and the maximum fiscal-rule residual is 4.91×10^{-5} . Capital, debt, the wedge, and informality have at most one direction reversal; a two-cycle is not an admissible solution.

5.4 Transition to a fully funded formal system

Figure 2 repeats the experiment with the full 16 percent contribution credited to individual accounts. New cohorts earn no PAYG rights, but the government honors Law 100 benefits already accrued and retains the solidarity benefit.

The contrast is immediate. Informality falls to 45.3 percent for cohort 1, 23.5 percent for cohort 2, and 10.2 percent in the steady state. Capital rises monotonically after the predetermined first cohort, reaching 0.0452 for cohort 2 and 0.0611 in the limit. The fiscal wedge falls through zero between cohorts 5 and 10. Debt peaks at 58.1 percent of annual GDP and returns to the same 55 percent anchor. Aggregate welfare rises 2.7 percent for cohort 1, 9.0 percent for cohort 2, and 22.9 percent in the stationary allocation. The terminal and maximum fiscal-rule gaps are 2.04×10^{-4} and 4.00×10^{-6} .

5.5 Wage and welfare distributions

Figures 4–7 show that aggregate paths conceal large within-cohort differences. Under pillars, the first reform cohort loses 1.26 percent in aggregate, but its tenth percentile

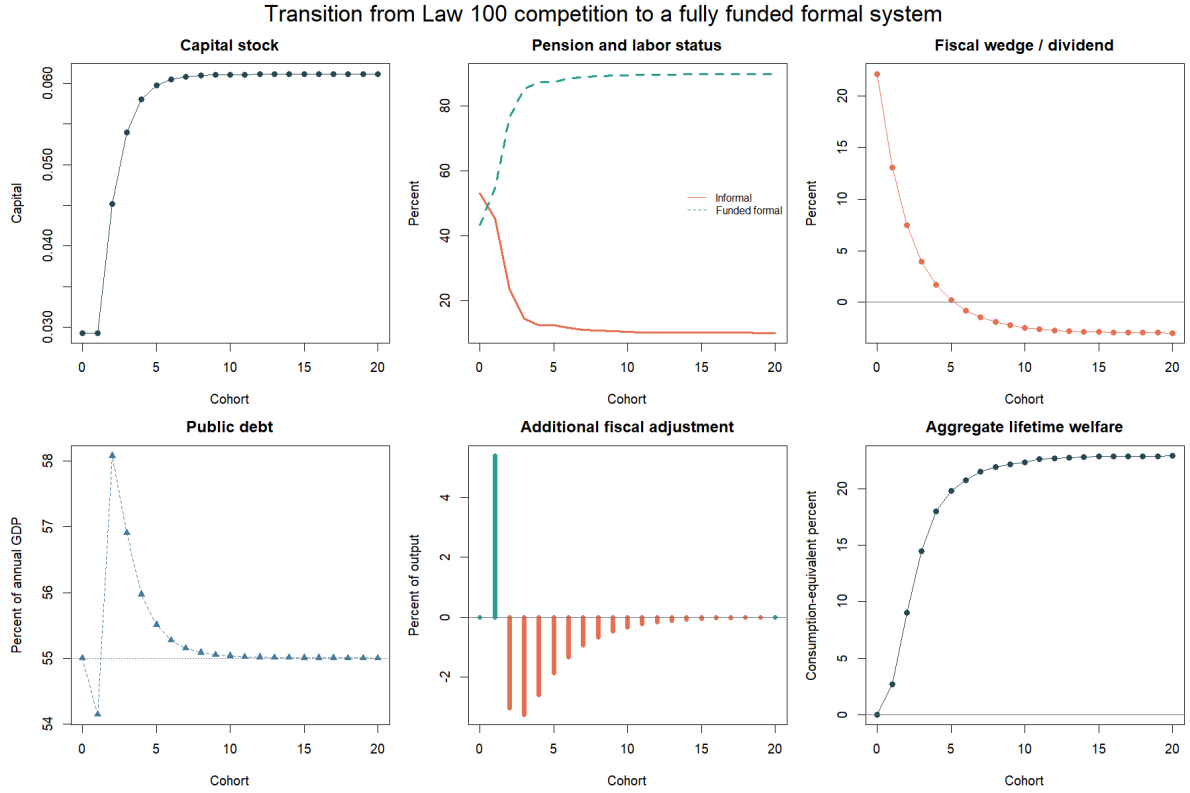


Figure 2: Transition from Law 100 to a fully funded formal system
Notes: A negative fiscal wedge is reported as a model-implied dividend. It does not overturn consumption taxes or other statutory taxes outside the model.

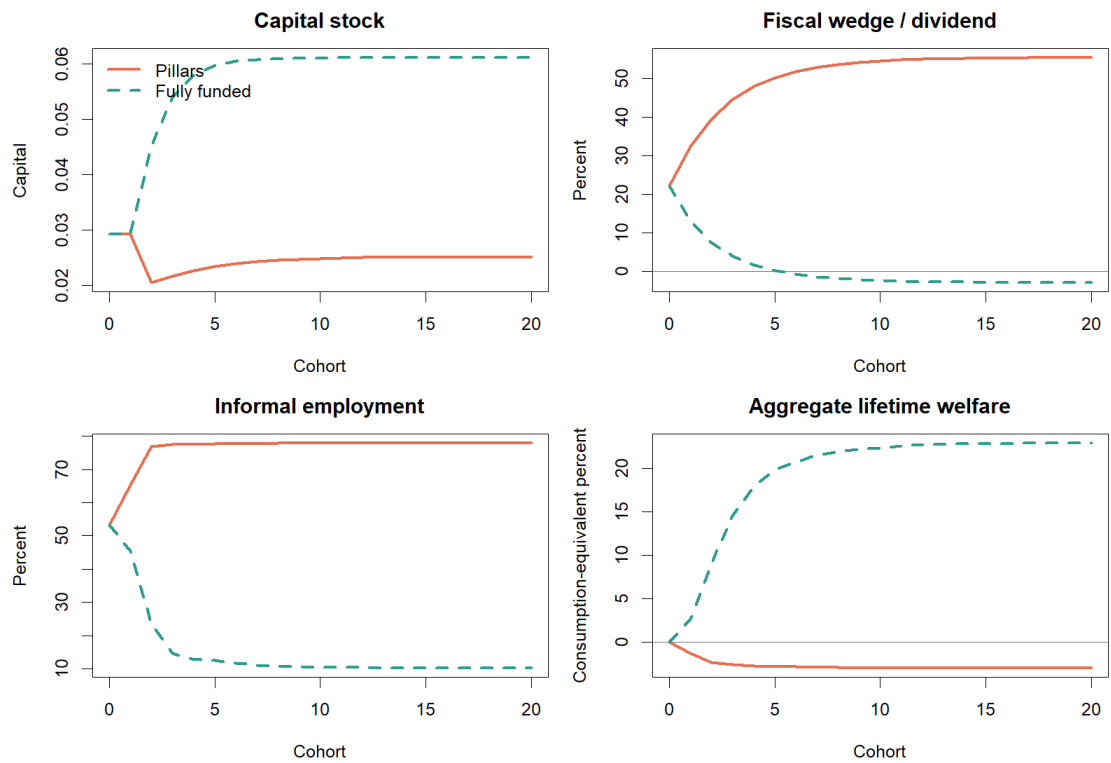


Figure 3: Pillars and full funding: transition comparison

loses 12.1 percent and its ninetieth percentile gains 2.4 percent. At the pillar endpoint, aggregate welfare is 2.99 percent below Law 100; 12.2 percent of types gain, the tenth percentile loses 20.5 percent, and the ninetieth percentile gains 1.5 percent. Those already old at the pillar reform lose 3.6 percent in aggregate even though the upper tail benefits from the solidarity transfer.



Figure 4: Distribution of labor earnings during the pillar transition

The fully funded counterfactual is also heterogeneous early in the transition. Cohort 1 gains 2.7 percent in aggregate, but 32.8 percent of types lose slightly. By cohort 5 every type gains. At the endpoint, the aggregate gain is 22.9 percent, the tenth percentile gains 7.9 percent, and the median gains 52.6 percent. Figure 7 compares stationary earnings distributions and type-level welfare profiles directly. The result is not a Pareto theorem: it is conditional on the model's contribution valuation, fiscal incidence, and omitted transition costs.

6 Conclusion

Moving from competing regimes to pillars changes more than pension assignment. Under Law 100, informality, capitalization, and PAYG coexist because portability and eligibility

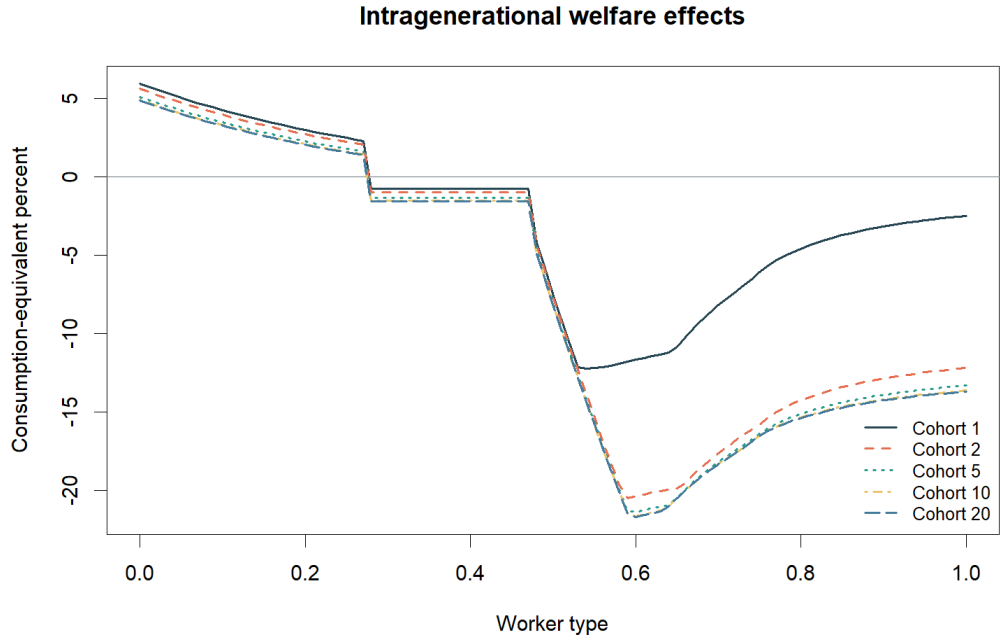


Figure 5: Intragenerational welfare during the pillar transition

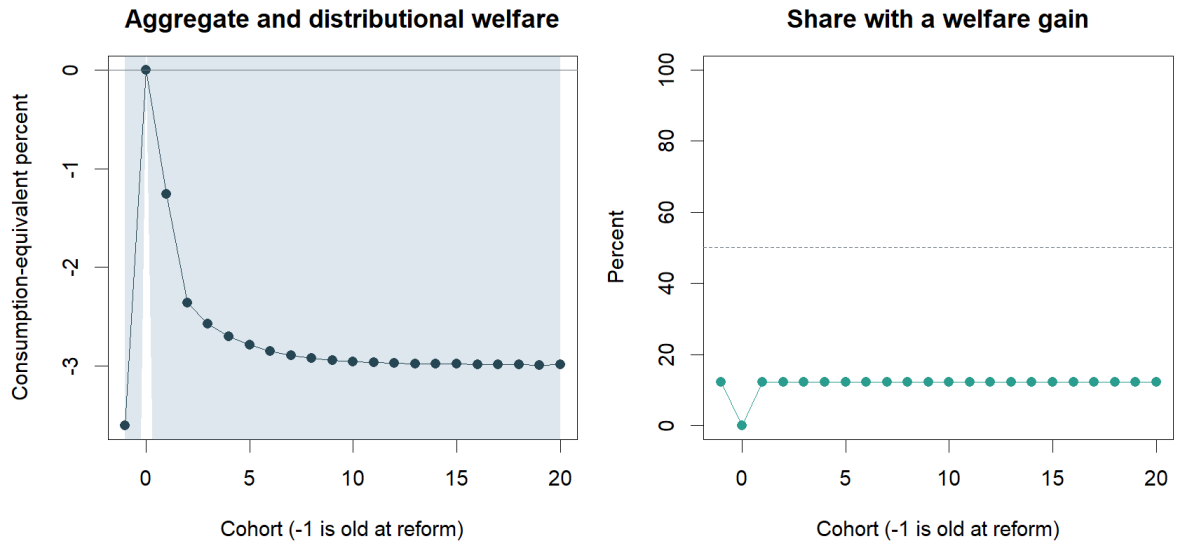


Figure 6: Pillar transition: intergenerational welfare and share of winners
Notes: Welfare uses experienced utility. The shaded area spans the tenth to ninetieth percentile of individual consumption-equivalent changes.

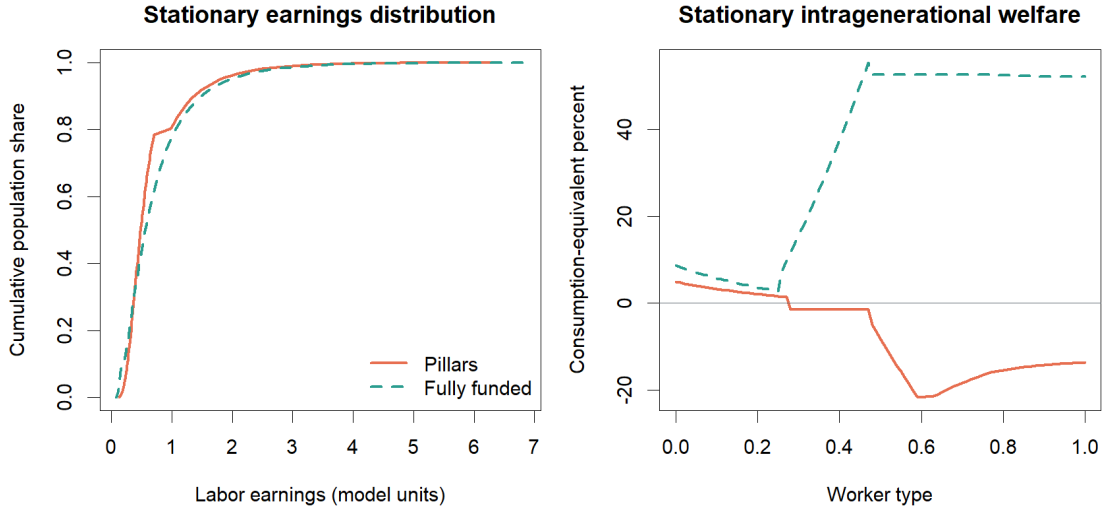


Figure 7: Stationary earnings and intragenerational welfare

create comparative advantage across worker types. Under Law 2381, PAYG becomes mandatory up to 2.3 minimum wages and only higher earnings generate funded balances. In the model, that design reduces funded saving substantially, raises informality, and contracts both output and the fiscal base.

The decomposition matters. The pillar steady state does not eliminate voluntary saving and it does not enlarge the net PAYG cash deficit. Voluntary saving almost triples and the PAYG deficit falls. Nevertheless, funded capital falls 70 percent, total capital falls 14 percent, and the consumption-tax base falls 44 percent as informality rises. The 55 percent stationary tax is therefore an equilibrium feedback among the composition of saving, formality, and the fiscal base. Treating it as the direct accounting cost of PAYG would be incorrect.

The two-instrument fiscal closure removes the former numerical accordion. The consumption wedge follows a smooth bounded path, while a temporary primary-balance instrument stabilizes public debt at the legal 55 percent anchor. Debt, taxes, labor status, and capital converge in both reforms, and the additional adjustment vanishes at each endpoint. This is a disciplined transition closure, but it does not identify which expenditure cuts or non-distortionary revenues would implement A_t in practice.

The fully funded counterfactual reverses the central pillar mechanisms: capital more than doubles, informality falls to 10 percent, and stationary welfare rises 23 percent. Those large gains are informative about the model's mechanism, not a policy ranking that can be exported unchanged. The experiment omits administrative fees, annuity-

market risk, portfolio choice, minimum-return guarantees, and a detailed financing plan for legacy obligations. Its residual fiscal dividend should be read as unused fiscal space inside the model.

Five cautions remain. First, Law 2381 has not entered into force, so the pillar exercise is counterfactual. Second, one period is about 40 years; the paths are generational, not annual forecasts. Third, 55 percent is the statutory long-run debt anchor, not current debt, and the 2026–2027 escape clause remains active. Fourth, setting $\omega_B = 0$ prevents mechanical domestic crowding out and is an identifying assumption. Fifth, the benefit multiplier, minimum-wage normalization, contribution histories, and the transition adjustment are not identified by the available calibration source.

The next step is a five-year multi-cohort model with age, sex, contribution histories, semicontributory rules, debt maturity, and an explicit menu of fiscal instruments. That extension would show whether the qualitative comparison survives with annual legal flows, legacy PAYG liabilities, and empirically calibrated domestic absorption. The present model’s main contribution is narrower: it separates pension design from an unstable fiscal closure and makes the saving, informality, tax-base, and welfare mechanisms auditable.

A Numerical algorithm

A.1 Household alternatives

For each candidate aggregate state (K, N_f, N_s, τ_c) , the algorithm computes prices and solves household problems at every diagnostic type. Under Law 100 the alternatives are informality, formal funded employment, and formal PAYG employment. Under pillars, the two choices are informality and the statutory PAYG–funded package. Under the fully funded counterfactual, the formal package sends the full contribution to the individual account. Each problem imposes nonnegative voluntary saving and positive consumption. Choice maximizes decision utility, while welfare uses experienced utility.

A.2 Locating endogenous thresholds

The diagnostic grid identifies every interval in which the selected alternative or pension classification changes. A scalar root finder solves the relevant utility equality within each bracket. Roots and the statutory pillar-income threshold are added to the quadrature

nodes. Choice is evaluated at interval midpoints and held fixed only inside those root-bounded intervals. The method neither assumes the number of crossings nor imposes their order.

A.3 Stationary general equilibrium

Given the exact intervals, trapezoidal quadrature aggregates labor, consumption, voluntary saving, funded contributions, PAYG flows, and welfare. The update map is

$$(K, N_f, N_s, \tau_c) \mapsto (\widehat{K}, \widehat{N}_f, \widehat{N}_s, \widehat{\tau}_c). \quad (39)$$

At each candidate allocation, output maps the annual legal targets into $\bar{B} = 0.55Y/H$ and $\bar{PB} = 0.002Y$. Gross household assets are converted to productive capital using (29), and the stationary fiscal wedge solves the consolidated budget including the primary-surplus target. Damped updates continue until the four state variables and fixed-point residuals satisfy tolerance.

The publication script solves 101 types and uses that allocation to initialize 201 types. Validation requires positive consumption, nonnegative saving, normalized regime shares, regime-appropriate threshold structure, the income identity, the debt anchor, the primary balance target, and maximum fixed-point residual below 5×10^{-5} . The four reported states—Law 100, central pillars, pillar stress, and full funding—all pass.

A.4 Transition path iteration

Let $T = 20$. The initial capital stock and public debt are predetermined by the Law 100 steady state, including $HB_0/Y_0 = 0.55$. From cohort 1 onward, entrants face either pillar or fully funded parameters. Those old at announcement retain their Law 100 contributory rights and accumulated saving.

For a candidate sequence

$$\{K_t, N_{ft}, N_{st}, \tau_{ct}\}_{t=0}^T,$$

the algorithm performs a backward-looking household/forward fiscal sweep:

1. compute current prices and the candidate next-period return;
2. solve each cohort's household problem and integrate gross assets and labor;
3. update the fiscal wedge through (35);

4. compute PB_t^{raw} , PB_t^* , and A_t using (36)–(37);
5. advance debt using (33) and map gross assets into capital using (29).

Debt is generated by the forward sweep rather than guessed independently. A positive A_t is an additional primary adjustment; the endpoint restriction is $A_0 = A_T = 0$.

Real paths are updated with an old-iterate weight of 0.75 and fiscal paths with a weight of 0.50. These are numerical relaxation parameters and do not alter the fixed point. A saved validated path may be used as a warm start, but every equation is reevaluated. Iteration stops only when the maximum state change is below 10^{-4} and the fiscal-rule residual below 2×10^{-4} .

The joint terminal gap uses an absolute fiscal-wedge error because the fully funded target is close to zero:

$$\Delta_T = \max \left\{ \frac{|\widehat{K}_T - \bar{K}|}{\bar{K}}, \frac{|\widehat{B}_T - \bar{B}|}{\bar{K}}, |\widehat{\tau}_{c,T-1} - \bar{\tau}_c| \right\}. \quad (40)$$

A path is accepted only if the internal fixed point converges, $\Delta_T < 10^{-3}$, and the maximum debt-identity and fiscal-rule residuals are below 2×10^{-4} . As an additional anti-accordion check, capital, debt, the wedge, and informality may have no more than two direction reversals. The pillar path has one; the fully funded path has none. Their terminal gaps are 4.00×10^{-4} and 2.04×10^{-4} , respectively. Debt-identity residuals are zero to machine precision, and maximum fiscal-rule residuals are 4.91×10^{-5} and 4.00×10^{-6} .

A.5 Distributional accounting

The code retains 101 type-level observations for every transition cohort. Population weights are quadrature weights from $f(i)$. Reports include weighted wage quantiles, Gini coefficients, winner shares, and welfare percentiles. Normative welfare uses experienced utility W , not the present-biased decision utility V . For $\sigma \neq 1$, the consumption-equivalent change under reform r relative to Law 100 is

$$\lambda_r(i) = \left[\frac{W^r(i)}{W^{100}(i)} \right]^{1/(1-\sigma)} - 1. \quad (41)$$

Those old at reform receive an old-age-only equivalent; young cohorts receive a lifetime equivalent. Aggregate equivalents are computed from weighted aggregate utilities and

are not the arithmetic mean of individual equivalents.

References

- Becerra, O. (2025, September). A simulation model for the Colombian pension system. Documento CEDE 2025-31, CEDE, Universidad de los Andes.
- Bobba, M., L. Flabbi, S. Levy, and M. Tejada (2021). Labor market search, informality, and on-the-job human capital accumulation. *Journal of Econometrics* 223(2), 433–453.
- Congreso de Colombia (1993). Ley 100 de 1993: Sistema de seguridad social integral. Diario Oficial No. 41.148.
- Congreso de Colombia (2021). Ley 2155 de 2021: Regla fiscal, ancla de deuda y limite de deuda. Artículo 60.
- Congreso de Colombia (2024). Ley 2381 de 2024: Sistema de proteccion social integral para la vejez, invalidez y muerte de origen comun. Diario Oficial No. 52.819.
- Corte Constitucional de Colombia (2025, June). Auto a-841 de 2025, expediente d-15.989.
- Departamento Administrativo Nacional de Estadística (2025). Pobreza monetaria en colombia: Año 2023. Technical report, DANE.
- Diamond, P. A. (1977). A framework for social security analysis. *Journal of Public Economics* 8(3), 275–298.
- Doligalski, P. and L. E. Rojas (2023). Optimal redistribution with a shadow economy. *Theoretical Economics* 18, 749–791.
- Feldstein, M. (1985). The optimal level of social security benefits. *The Quarterly Journal of Economics* 100(2), 303–320.
- Ferreira, P. C. and R. M. Parente (2020). Social security reform, retirement and occupational behavior. Working paper.
- İmrohoroglu, A., S. İmrohoroglu, and D. H. Joines (2003). Time-inconsistent preferences and social security. *The Quarterly Journal of Economics* 118(2), 745–784.

- Joubert, C. (2015). Pension design with a large informal labor market: Evidence from Chile. *International Economic Review* 56(2), 673–694.
- Laibson, D. (1998). Life-cycle consumption and hyperbolic discount functions. *European Economic Review* 42(3–5), 861–871.
- Levy, S. (2008). *Good Intentions, Bad Outcomes: Social Policy, Informality, and Economic Growth in Mexico*. Brookings Institution Press.
- McKiernan, K. (2020). Social security reform in the presence of informality. Manuscript, Review of Economic Dynamics.
- Ministerio de Hacienda y Credito Publico (2026). Marco fiscal de mediano plazo 2026. Technical report, Republica de Colombia, Bogota.
- Moreno, C. (2022). The impact of pension systems in labor markets with informality. Working paper.
- Pardo, O. (2023). Mandatory retirement savings in the presence of an informal labor market. *Journal of Population Economics* 36, 2857–2888.
- Presidencia de la Republica de Colombia (2022). Decreto 2613 de 2022: Salario minimo mensual legal para 2023.
- Ulyssea, G. (2018). Firms, informality, and development: Theory and evidence from Brazil. *American Economic Review* 108(8), 2015–2047.